

(ii) If X and Y are topological spaces and $f: X \rightarrow Y$ is continuous, then f is Borel measurable.

In fact, by (i) we consider $(X, \beta(X))$ and $(Y, \beta(Y))$, and by (ic) f is Borel measurable if it is measurable w.r.t. these Borel σ -algebras.

By (i) (and Corollary 8.3) this follows from the fact that if $U \subset Y$ is open then $f^{-1}(U) \subset X$ is open, hence Borel.

In particular, $\forall F \in \beta(Y)$, $f^{-1}(F) \in \beta(X)$.

[see Example 4.12, Fact #3]

end local #1

Examples 8.5

(i) let $f: \mathbb{R} \rightarrow \mathbb{R}$ be an increasing function.

We claim that f is Borel measurable (see Remark 8.4 (ii)).

In view of Corollary 8.3, it suffices to show that

for all $a \in \mathbb{R}$, $f^{-1}((a, +\infty)) \in \mathcal{B}(\mathbb{R})$.

Since f is increasing, if $b := \inf f^{-1}((a, +\infty))$ then

either $f^{-1}((a, +\infty)) = (b, +\infty) \in \mathcal{B}(\mathbb{R})$

or $f^{-1}((a, +\infty)) = [b, +\infty) \in \mathcal{B}(\mathbb{R})$.

(ic) let $(X, \mathcal{M})$ be a measurable space, and let $E \subset X$.

The characteristic function of E is defined by

$$\chi_E: X \rightarrow \mathbb{R}, \quad \chi_E(x) = \begin{cases} 1 & \text{if } x \in E, \\ 0 & \text{if } x \notin E. \end{cases}$$

$$\text{If } B \subseteq \mathbb{R} \text{ then } \chi_E^{-1}(B) = \begin{cases} E & \text{if } 1 \in B, 0 \notin B, \\ X \setminus E & \text{if } 0 \in B, 1 \notin B, \\ X & \text{if } 0 \in B, 1 \in B, \\ \emptyset & \text{if } 0 \notin B, 1 \notin B. \end{cases}$$

Thus χ_E is measurable iff $E \in \mathcal{M}$.

Proposition 8.6

let $(X, \mathcal{M})$, $(Y, \mathcal{N})$, $(Z, \mathcal{D})$ be measurable spaces, let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ measurable functions.

Then $g \circ f: X \rightarrow Z$ is measurable.

Proof If $F \in \mathcal{D}$ then $g^{-1}(F) \in \mathcal{N}$ and so $(g \circ f)^{-1}(F) = f^{-1}(g^{-1}(F)) \in \mathcal{M}$. ■

Remark 8.7

Lead (i) If $(X, \mathcal{M})$ is a measurable space, $f: X \rightarrow \mathbb{R}^d$ is measurable and $g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is Borel measurable, then $g \circ f$ is measurable.

$$f: X \rightarrow \mathbb{R}^d \text{ meas.} \\ \downarrow \beta(\mathbb{R}^d)$$

$$g: \mathbb{R}^d \rightarrow \mathbb{R}^d \text{ Borel meas.} \\ \downarrow \beta(\mathbb{R}^d) \downarrow \beta(\mathbb{R}^d)$$

$$\Downarrow \\ g \circ f \text{ meas.}$$

This follows from Example 8.4 (i), per which for $f: X \rightarrow \mathbb{R}^d$ we consider the target as $(\mathbb{R}^d, \beta(\mathbb{R}^d))$.

Also, with $g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ we take its target as $(\mathbb{R}^d, \beta(\mathbb{R}^d))$.

Therefore, if $B \in \beta(\mathbb{R}^d)$ then $g^{-1}(B) \in \beta(\mathbb{R}^d)$ (g is Borel meas.)
 $\Rightarrow f^{-1}(g^{-1}(B)) \in \mathcal{M}$, i.e., $(g \circ f)^{-1}(B) \in \mathcal{M}$.

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ Borel meas.} \\ \downarrow \beta(\mathbb{R}) \downarrow \beta(\mathbb{R})$$

$$g: \mathbb{R} \rightarrow \mathbb{R} \text{ Lebesgue meas.} \\ \downarrow \beta(\mathbb{R}) \downarrow \beta(\mathbb{R})$$



$g \circ f$ meas.

(ii) let $\tilde{f}, \tilde{g}: \mathbb{R} \rightarrow \mathbb{R}$ be 'such that $\tilde{g}$ is Lebesgue measurable (i.e., $g: (\mathbb{R}, \mathcal{L}^1) \rightarrow (\mathbb{R}, \beta(\mathbb{R}))$) and

$\tilde{f}$ is Borel measurable (i.e., $f: (\mathbb{R}, \beta(\mathbb{R})) \rightarrow (\mathbb{R}, \beta(\mathbb{R}))$).

Then $\tilde{g} \circ \tilde{f}$ may fail to be measurable,
 $\tilde{g} \circ \tilde{f}: (\mathbb{R}, \beta(\mathbb{R})) \rightarrow (\mathbb{R}, \beta(\mathbb{R}))$.

Indeed, by Example 4.12 there exists a set $F \subseteq \mathbb{R}$ that is Lebesgue measurable and not Borel measurable, with $F = \tilde{g}^{-1}(E)$, $\tilde{g}$ continuous, $\tilde{g}^{-1}$ continuous, and E is not Lebesgue measurable.

Set $\tilde{f} := \bar{g}^{-1}$, continuous hence Borel measurable by Rmk 8.4(iv),
 and set $\tilde{g} := \chi_F$, Lebesgue measurable in view of Example 8.5(ii).
 then $\{1\} \in \beta(\mathbb{R})$ and
 $(\tilde{g} \circ \tilde{f})^{-1}(\{1\}) = \tilde{f}^{-1}(\tilde{g}^{-1}(\{1\})) = \tilde{f}^{-1}(F) = g^{-1}(F) = E$
 which is not Lebesgue measurable, so $\tilde{g} \circ \tilde{f}$ is not measurable

Proposition 8.8 let $(X, \mathcal{M})$ be a measurable space, and let
 $f, g : X \rightarrow [-\infty, +\infty]$ be arbitrary. Then

f, g are measurable iff $(f, g) : X \rightarrow [-\infty, +\infty]^2$ is measurable.

Proof $\Rightarrow$ Suppose that f and g are measurable.

If $U, V \subseteq [-\infty, +\infty]$ are open, then

$$(f, g)^{-1}(U \times V) = f^{-1}(U) \cap g^{-1}(V) \in \mathcal{M}.$$

In particular, if $R \subseteq [-\infty, +\infty]^2$ is an open rectangle then
 (recall, $Z \subseteq [-\infty, +\infty]^2$ is open iff $Z \cap (-\infty, +\infty)^2$ is open)

$(f, g)^{-1}(R) \in \mathcal{M}$, and, by Corollary 8.3, (f, g) is measurable.

$\Leftarrow$ let $\pi_1 : [-\infty, +\infty]^2 \rightarrow [-\infty, +\infty]$ and

$$(x, y) \mapsto x$$

$$\pi_2 : [-\infty, +\infty]^2 \rightarrow [-\infty, +\infty]$$

$$(x, y) \mapsto y$$

be the (continuous) canonical projections.

By Remark 8.4 (iv) $\pi_i, i=1,2$, are Borel measurable,
 so by Remark 8.7 (c)

$$f = \pi_1 \circ (f, g) \text{ and } g = \pi_2 \circ (f, g)$$

are measurable functions.

Corollary 8.9

Let $(X, \mathcal{M})$ be a measurable space, and let $f: X \rightarrow \mathbb{R}$ be measurable. Then so are

f^2 , $|f|$, f^+ , f^- , αf ($\alpha \in \mathbb{R}$), with $f^+ := \max\{f, 0\}$, $f^- := \max\{-f, 0\}$.
Furthermore, if $f, g: X \rightarrow \mathbb{R}$ are measurable then so are $f \pm g$, $f \cdot g$, $\min\{f, g\}$, $\max\{f, g\}$.

Proof: The first part follows from Remark 8.7 (i), and the second part also uses Prop 8.8.

For example, if f, g are measurable then (f, g) is measurable, with $f+g = \theta \circ (f, g)$ where θ is the continuous (hence, Borel) function

$$\theta: \mathbb{R}^2 \rightarrow \mathbb{R}, (x, y) \mapsto x+y. \quad \blacksquare$$

Remark 8.10 There are versions of Corollary 8.9 applicable to functions having values in the extended real line $\overline{\mathbb{R}} := [-\infty, +\infty]$, provided the operations are well defined.

For example, for $f+g$ need $(f(x), g(x)) \notin \{\pm(\infty, -\infty)\}$ for all $x \in X$. Concerning $f \cdot g$, define $(f \cdot g)(x) = 0$ whenever $f(x)$ or $g(x)$ is zero.

Proposition 8.11 Let $(X, \mathcal{M})$ be a measurable space, and let $f_n: X \rightarrow [-\infty, +\infty]$ be measurable functions.

Then $\sup_n f_n$, $\inf_n f_n$, $\liminf_n f_n$, $\limsup_n f_n$ are measurable.

In particular, if $f_n \rightarrow f$ pointwise, i.e. for every $x \in X$ $f(x) = \lim_{n \rightarrow \infty} f_n(x)$, then f is measurable.

Proof For every $a \in \mathbb{R}$,
 $(\sup_n f_n)^{-1}((a, +\infty]) = \bigcup_{n=1}^{\infty} f_n^{-1}((a, +\infty]) \in \mathcal{M}$,
 and by Corollary 8.3 and Remark 8.4(ii) it follows that
 $\sup_n f_n$ is measurable. (8.1)

likewise, $(\inf_n f_n)^{-1}([-\infty, a)) = \bigcup_{n=1}^{\infty} f_n^{-1}([-\infty, a)) \in \mathcal{M}$,

and again by Corollary 8.3, we have that
 $\inf_n f_n$ is measurable (8.2)

[Alternatively: by Remark 8.10 and Corollary 8.9,

- f_n are measurable, so by (B.1)

$\sup_n (-f_n)$ is measurable,

and again by Remark 8.10 and Corollary 8.9, we deduce that
 $\inf_n f_n = -\sup_n (-f_n)$ is measurable].

Now, $\limsup_{n \rightarrow +\infty} f_n = \inf_{n \in \mathbb{N}} \left(\sup_{k > n} f_k \right)$ which is

measurable by (8.1) and (8.2), and similarly,

$\liminf_{n \rightarrow +\infty} f_n = \sup_{n \in \mathbb{N}} \left(\inf_{k > n} f_k \right)$ is measurable. ■

Def 8.12 Let $(X, \mathcal{M}, \mu)$ be a measure space. We say that a property P holds almost everywhere (a.e.) on a set $A \subseteq X$ if there is a set of measure zero, $E \in \mathcal{M}$, such that P holds on $A \setminus E$. We say that P holds a.e. if it holds a.e. on X .

Example 8.13

- (i) Almost every real number is irrational.
- (ii) The function $f(x) := |x|$, $x \in \mathbb{R}$, is differentiable a.e.
- (iii) Rademacher's Theorem Every Lipschitz function $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable a.e.

$$\exists L > 0 \forall x, y \quad |f(x) - f(y)| \leq L|x - y|$$

Proposition 8.14

Let $(X, \mathcal{M}, \mu)$ be a measure space with μ a complete measure, and let $(Y, \mathcal{N})$ be a measurable space.

Let $f: X \rightarrow Y$ be a measurable function, and let $g: X \rightarrow Y$ be such that $f = g$ a.e.

Then g is measurable.

Proof Let $E_0 := \{x \in X: f(x) \neq g(x)\}$.

By Def 8.12 $\exists E \in \mathcal{M}$ s.t. $\mu(E) = 0$ and $f(x) = g(x)$ if $x \in X \setminus E$.

Then $X \setminus E \subseteq X \setminus E_0$, hence

$$E_0 \subseteq E$$

and μ is complete (see Def. 4.8), therefore

$$E_0 \in \mathcal{M} \text{ and } \mu(E_0) = 0. \quad (8.2)$$

Let $F \in \mathcal{N}$. We have $? g^{-1}(F) \in \mathcal{M}?$

$$\begin{aligned} g^{-1}(F) &= \{x \in X \setminus E_0: g(x) \in F\} \cup \{x \in E_0: g(x) \in F\} \\ &= \{x \in X \setminus E_0: f(x) \in F\} \cup \{x \in E_0: g(x) \in F\} \\ &= (f^{-1}(F) \setminus E_0) \cup \{x \in E_0: g(x) \in F\}. \end{aligned}$$

Now, $f^{-1}(F) \setminus E_0 \in \mathcal{M}$ (see (8.2)) and

$$\{x \in E_0: g(x) \in F\} \subseteq E_0,$$

thus it belongs to $\mathcal{M}$ by (8.2) and because μ is complete.

Corollary 8.15

let $(X, \mathcal{M}, \mu)$ be a measure space, with μ a complete measure. let $f_n: X \rightarrow [-\infty, +\infty]$ be measurable functions,

If there exists $\lim_{n \rightarrow \infty} f_n(x)$ for μ a.e. $x \in X$, then $\lim_{n \rightarrow \infty} f_n$ is measurable.

Proof Assume first that there exists $E \in \mathcal{M}$, $\mu(E) > 0$, such that $\lim_{n \rightarrow \infty} f_n(x)$ exists for all $x \in X \cap E$.

By Proposition 8.11

$f(x) := \lim_{n \rightarrow \infty} f_n(x)$ is measurable on $X \cap E$, so if we "define" $\lim_{n \rightarrow \infty} f_n$ as

$$g(x) := \begin{cases} \lim_{n \rightarrow \infty} f_n(x) & \text{if } x \in X \cap E \\ 0 & \text{otherwise} \end{cases}$$

(any other definition on E will do...), then by Prop 8.14 g is measurable. ■

Def 8.16

let $(X, \mathcal{M})$ be a measurable space. we say that $s: X \rightarrow \mathbb{R}$ is a simple function if it is measurable and has finite range, i.e.,

$$(*) \quad s = \sum_{i=1}^n a_i \chi_{E_i}, \quad n \in \mathbb{N},$$

(**) $a_i \neq a_j$ if $i \neq j$, $E_i := s^{-1}(\{a_i\}) \in \mathcal{M}$.

(*) + (**): standard representation $\downarrow$ on $\mathbb{R}$ consider $\beta(\mathbb{R})$
 s measurable $\Rightarrow E_i \in \mathcal{M}$

It is clear that if $f(x) = +\infty$ then $s_n(x) \uparrow f(x)$,
and if $f(x) = -\infty$ then $s_n(x) \downarrow f(x)$. Further, if $f(x) \in \mathbb{R}$
choose $n_0 \in \mathbb{N}$ such that $f(x) \in [-n_0, n_0]$.

If $n > n_0$, then $f(x) \in [-n, n]$ and

$$|s_n(x) - f(x)| \leq \frac{1}{2^n} \xrightarrow{n \rightarrow \infty} 0.$$

The latter observation also proves (iii). ■

end of #12

Prop 8.18 let $(X, \mathcal{M}, \mu)$ be a measure space, and
let $(X, \mathcal{M}_\mu, \mu)$ be its completion.

Then $f: X \rightarrow \mathbb{R}$ is $\mathcal{M}_\mu$ -measurable iff
there exists $g: X \rightarrow \mathbb{R}$ $\mathcal{M}$ -measurable such that
 $f = g$ a.e. (wrt μ).

Proof [$\Leftarrow$] let $U \subseteq \mathbb{R}$ be an open set, and let
 $E := \{x \in X: f(x) \neq g(x)\}$

be such that $E \in \mathcal{M}$ and $\mu(E) = 0$. Thus

$$\begin{aligned} f^{-1}(U) &= \{x \notin E: f(x) \in U\} \cup \{x \in E: f(x) \in U\} \\ &= \{x \in X \setminus E: g(x) \in U\} \cup \{x \in E: f(x) \in U\} \\ &= g^{-1}(U) \cap (X \setminus E) \cup \{x \in E: f(x) \in U\}. \end{aligned}$$

Now, $g^{-1}(U) \cap (X \setminus E) \in \mathcal{M} \subset \mathcal{M}_\mu$ (see Def 7.1)

and $\{x \in E: f(x) \in U\} \subset E \in \mathcal{M}$, with $\mu(E) = 0$, i.e.,

$\{x \in E: f(x) \in U\} \in \mathcal{N}$ (see before Def. 7.1)

with $\mathcal{N} \subset \mathcal{M}_\mu$.

Since $\mathcal{M}_\mu$ is a σ -algebra (see Thm 7.4), we obtain

$$f^{-1}(U) \in \mathcal{M}_\mu.$$

[$\Rightarrow$] Step 1 Suppose that $f = \chi_E$, with $E \in \mathcal{M}_\mu$.

Then $E = B \cup A$, $B \in \mathcal{M}$, $A \in \mathcal{N}$ (see Def 7.1), i.e., there exist $H \in \mathcal{M}$ with $\mu(H) = 0$ and $A \subseteq H$.

Let $g := \chi_{B \cup H}$, which is $\mathcal{M}$ -measurable.

claim if $x \notin H$ then $f(x) = g(x)$.

Indeed, either $x \notin E$ and then $f(x) = g(x) = 0$, or $x \in E$ and $x \notin H \Rightarrow x \in E \setminus A = B$ with $x \notin H \Rightarrow f(x) = 1 = g(x)$.

Step 2 If s is a simple $\mathcal{M}_\mu$ -measurable function, then

$$s = \sum_{i=1}^n a_i \chi_{E_i}, \quad a_i \in \mathbb{R}, E_i \in \mathcal{M}_\mu,$$

and by step 1 $s = \tilde{s}$ μ -a.e., with

$$\tilde{s} := \sum_{i=1}^n a_i \chi_{\tilde{E}_i}, \quad \tilde{E}_i \in \mathcal{M},$$

$$\chi_{\tilde{E}_i} = \chi_{E_i} \quad \text{if } x \in X \setminus H_i, \quad H_i \in \mathcal{M}, \mu(H_i) = 0.$$

Therefore, $s = \tilde{s}$ if $x \in X \setminus (\bigcup_{i=1}^n H_i)$, where $\bigcup_{i=1}^n H_i \in \mathcal{M}$, $\mu(\bigcup_{i=1}^n H_i) = 0$.

Step 3 for an arbitrary f $\mathcal{M}_\mu$ -measurable, by Thm 8.17

let $s_n: X \rightarrow \mathbb{R}$ be $\mathcal{M}_\mu$ -measurable such that $s_n(x) \rightarrow f(x) \quad \forall x \in X$.